

MURALI SAI BUDDAKKAGARI VENKATA

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SUMMARY

AI Engineer specializing in production-grade AI agents, LLMs, retrieval-augmented generation, and multimodal multi-agent systems. Proficient in Python, LangChain, LlamaIndex, HuggingFace, and building scalable ML pipelines on GCP and AWS. Experienced translating business requirements into technical implementations and iterating on model strengths and inefficiencies to optimize AI agents for enterprise deployment.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C++, PySpark
- **AI & Machine Learning:** Prompt Engineering, Model Evaluation & LLM Evaluation (RAGAS), Retrieval-Augmented Generation, Multi-Agent Systems, Model Deployment, CrewAI, HuggingFace, Transformers, PyTorch, TensorFlow, Model Context Protocol (MCP), LangChain, LangGraph, LlamaIndex, FastAPI, Embedding Models, Scikit-learn, ChromaDB, Modern Large Language Models (LLMs), Modern AI Frameworks and APIs, Generative AI, Data-driven Modeling
- **Cloud, DevOps & Infrastructure:** AWS Deployments, Google Cloud Platform, BigQuery, Azure Services, Docker, PostgreSQL, Relational Databases, GitHub Actions, CI/CD, ReactJS, OpenAI API, Claude API

EXPERIENCE

Querkey Inc **Jun 2025 - Aug 2025**
Data Science Intern *CA, US*

- Built end-to-end data-driven modeling and machine learning ingestion pipelines using Python, PySpark, GCP, BigQuery, and scikit-learn, automating the processing of 200+ MP4 files daily and reducing validation latency by 40% to enable data-driven operational decisions at scale
- Deployed Streamlit and Superset-based real-time observability stack over 100K+ records with Dockerized MLflow model tracking, HuggingFace model monitoring, ChromaDB integration, and FastAPI endpoints on AWS infrastructure, improving production ML workflow reliability by 60% and enabling proactive incident response for business-critical pipelines

VIEW Synergy **Jan 2024 - Apr 2024**
AI and Automation Intern *KA, IND*

- Architected and deployed generative AI-driven workflow automation pipelines integrating ReactJS, TypeScript, RESTful APIs, Azure cloud services, LangChain, and LlamaIndex, enabling intelligent task orchestration and reducing manual process overhead for 100+ employees across full SDLC delivery cycles
- Developed retrieval-augmented generation microservices-based backend systems using C#, .NET 8.0, SQL, Docker, and the OpenAI API, delivering production-grade, low-latency API serving within an Agile SCRUM delivery model and accelerating cross-functional release cycles

PROJECTS

- RAG Enterprise - SEC EDGAR Filing Analyzer**
- Built a production RAG system over SEC 10-K filings using Python, LangChain, LlamaIndex, BeautifulSoup, and ChromaDB to meet client requirements, improving faithfulness (0.57→0.65), answer relevancy (0.58→0.68), and context recall (0.57→0.70) by evaluating model strengths and retrieval inefficiencies via RAGAS with GPT-4o-mini.
 - Enforced investment-banking-grade RBAC, MNPI guardrails, and SEC 17a-4 / FINRA 4511-aligned audit trails at the vector-store layer; managed production code and automated deployments via Docker, Terraform-provisioned AWS ECS Fargate, GitHub Actions CI/CD, and performed end-to-end code debugging.

- Multi-Agent Marketing Intelligence System**
- Gathered marketing business requirements and translated them into a Python-based multi-agent system with five specialized agents (research, analysis, content generation, critique, orchestration), orchestrating asynchronous tasks via CrewAI and Gemini API and integrating LangChain for robust agent coordination.
 - Developed end-to-end data pipelines on AWS (S3, Lambda) and PostgreSQL, applied software engineering best practices in CI/CD, and built Streamlit dashboards to visualize campaign performance-reducing analysis time by 60% and deploying secure real-time access via Streamlit Cloud.

- WorkFit**
- Developed a real-time multimodal posture monitoring system using CNN-based computer vision and flex sensor fusion, achieving 40% discomfort reduction and 85% user adoption, recognized with the Best Research Paper Award at ICICC 2024 for real-world impact and technical novelty.
 - Designed and trained a deep learning inference pipeline for simultaneous eye-strain and postural deviation detection, optimizing for low-latency real-time predictions on resource-constrained hardware.

EDUCATION

Stevens Institute of Technology **May 2026**
Master of Science, Applied Artificial Intelligence *NJ, US*

PES University **May 2024**
Bachelor of Technology, Computer Science (Specialization in ML) *KA, IND*